

# Required practical: I–V characteristics

Year 10 GCSE Physics • AQA Trilogy RP 16 • Physics RP 4

Name \_\_\_\_\_

Class \_\_\_\_\_ Date \_\_\_\_\_

Plan. Measure. Explain.

Work in pairs. Read the whole method before you build anything.

Online version with hints: [panphy.app/gcsephy/year10phy/unit01/practical/iv-characteristics/](https://panphy.app/gcsephy/year10phy/unit01/practical/iv-characteristics/)

## 1 Aim

Use circuit diagrams to construct circuits that investigate the I–V characteristics of a filament lamp, a diode and a resistor at constant temperature.

$$R = V \div I$$

R in ohms ( $\Omega$ ) · V in volts (V) · I in amperes (A)

## 2 Apparatus

- Variable d.c. power supply (0–6 V) or battery pack
- Variable resistor (rheostat)
- Ammeter (0–1 A) and milliammeter (0–100 mA), or multimeters
- Voltmeter (0–10 V)
- Switch and connecting leads
- Fixed resistor, for example 10  $\Omega$
- Filament lamp, for example 6 V 0.3 A
- Diode and a protective resistor, for example 100  $\Omega$

## 3 Safety

- The lamp gets hot. Do not touch it while it is on or just after.
- Do not go above the lamp's rated p.d. (for example 6 V).
- Always use the protective resistor with the diode, so the current stays small.

# Before you start

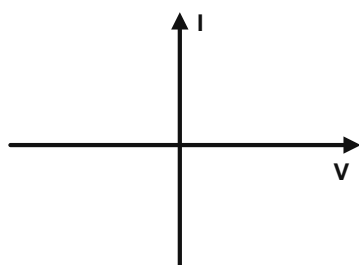
Learn the key words, then make a prediction before you build anything.

## 4 Key words

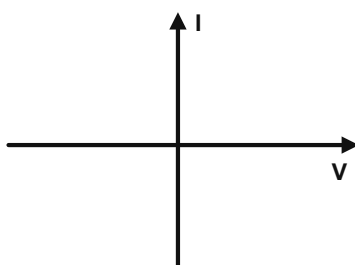
|                             |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| <b>Independent variable</b> | The variable you change.                                                                |
| <b>Dependent variable</b>   | The variable you measure.                                                               |
| <b>Control variable</b>     | A variable you keep the same, so the test is fair.                                      |
| <b>Resolution</b>           | The smallest change a meter can show, for example 0.01 A.                               |
| <b>Anomalous result</b>     | A result that does not fit the pattern of the others.                                   |
| <b>I–V characteristic</b>   | A graph of current (y-axis) against p.d. (x-axis) for a component.                      |
| <b>Ohmic conductor</b>      | A component whose current is directly proportional to the p.d. at constant temperature. |

## 5 Predict

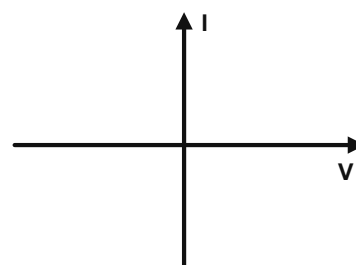
Sketch the I–V graph you expect for each component. Show both positive and negative p.d.



Fixed resistor



Filament lamp



Diode

# Method Set up and measure

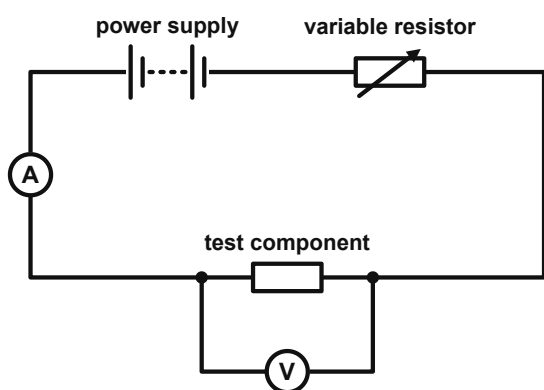
Method • I–V characteristics • Use the variable resistor to change the p.d. across each component. Record the current each time, then reverse the supply and repeat.

## 1 Variables

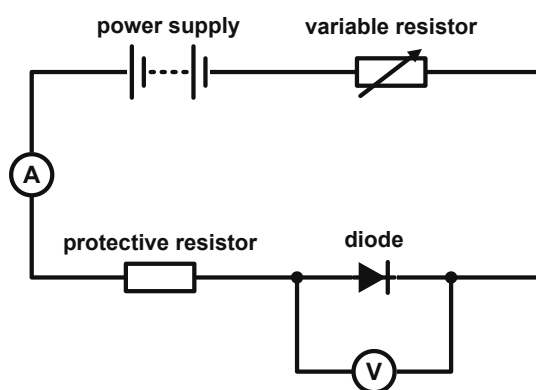
| Variable    | What it is                        | How                                         |
|-------------|-----------------------------------|---------------------------------------------|
| Independent | p.d. across the component         | Adjust the variable resistor or the supply  |
| Dependent   | Current through the component     | Read the ammeter                            |
| Control     | Temperature of the fixed resistor | Small currents; switch off between readings |

## 2 Method

1. Build the first circuit below with the fixed resistor as the test component. The voltmeter goes across the component only. ☐
2. Set the variable resistor to its highest resistance. Close the switch and record the p.d. and the current. ☐
3. Change the variable resistor to increase the p.d. in steps of about 1 V. Record the p.d. and current each time. Switch off between readings so the resistor stays cool. ☐
4. Swap the connections to the power supply so the current flows the other way. Repeat steps 2 and 3. Record these p.d. and current values as negative numbers. ☐
5. Replace the resistor with the filament lamp and repeat steps 2 to 4. Do not go above the lamp's rated p.d. ☐
6. Build the second circuit for the diode. Use a milliammeter and increase the p.d. in small steps of about 0.1 V from 0.4 V, where the current starts to rise quickly. Then reverse the supply and take readings. ☐



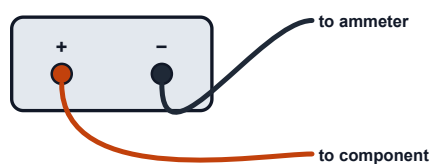
Test the resistor first, then swap in the filament lamp.



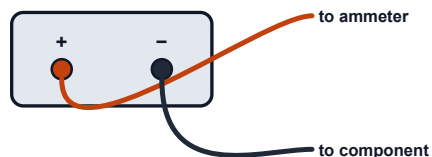
For the diode: add a protective resistor and use a milliammeter. The voltmeter goes across the diode only.

# Method Set up and measure

Forward: record + values



Reversed: swap the two leads



Swap the two leads at the power supply. The meters now show negative readings: write the minus sign.

real diode



symbol



It conducts when the current flows towards the band (the bar).

The band on the diode matches the bar in its symbol.  
Forward: current flows towards the band.

**Why reverse the supply?** An I–V graph shows both directions. Reversing the connections gives the negative half of the graph.

**Using the variable resistor.** Start at its highest resistance, so the p.d. across the component is small. Lower its resistance a step at a time to increase the p.d.

# Results Record your readings

Results • I–V characteristics • Record every reading as you take it. Reversed readings are negative: write the minus sign.

## 1 Results: fixed resistor

| Forward  |             | Reversed (negative) |             |
|----------|-------------|---------------------|-------------|
| p.d. / V | Current / A | p.d. / V            | Current / A |
|          |             |                     |             |
|          |             |                     |             |
|          |             |                     |             |
|          |             |                     |             |
|          |             |                     |             |
|          |             |                     |             |

## 2 Results: filament lamp

Calculate the resistance,  $R = V \div I$ , for each forward reading.

| Forward  |             |                       | Reversed (negative) |             |
|----------|-------------|-----------------------|---------------------|-------------|
| p.d. / V | Current / A | Resistance / $\Omega$ | p.d. / V            | Current / A |
|          |             |                       |                     |             |
|          |             |                       |                     |             |
|          |             |                       |                     |             |
|          |             |                       |                     |             |
|          |             |                       |                     |             |
|          |             |                       |                     |             |

# Results Record your readings

## 3 Results: diode

Use a milliammeter. Take small steps near 0.6 V.

| Forward  |              | Reversed (negative) |              |
|----------|--------------|---------------------|--------------|
| p.d. / V | Current / mA | p.d. / V            | Current / mA |
|          |              |                     |              |
|          |              |                     |              |
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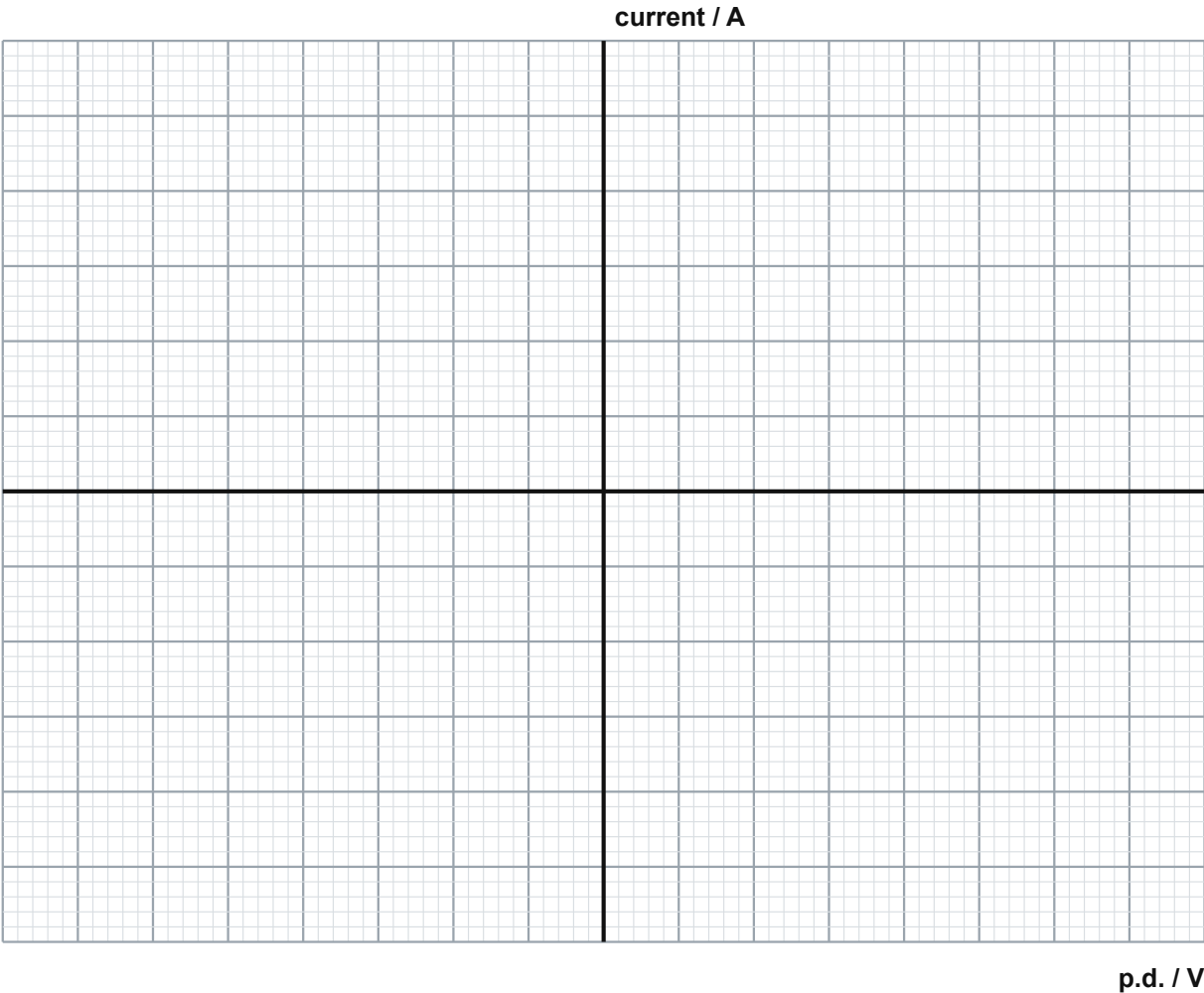
**Check your readings.** Every graph should pass through the origin: with no p.d. across a component, there is no current through it.

# Analysis    Graphs and conclusions

Analysis • I–V characteristics • Plot current (y-axis) against p.d. (x-axis) for each component, using the positive and negative readings. Then explain each shape.

## 1    Fixed resistor graph

Plot current against p.d. for the fixed resistor. Add scales to both axes. Draw a line of best fit.



**Q1** Describe the I–V graph for the fixed resistor. What does it tell you about its resistance?

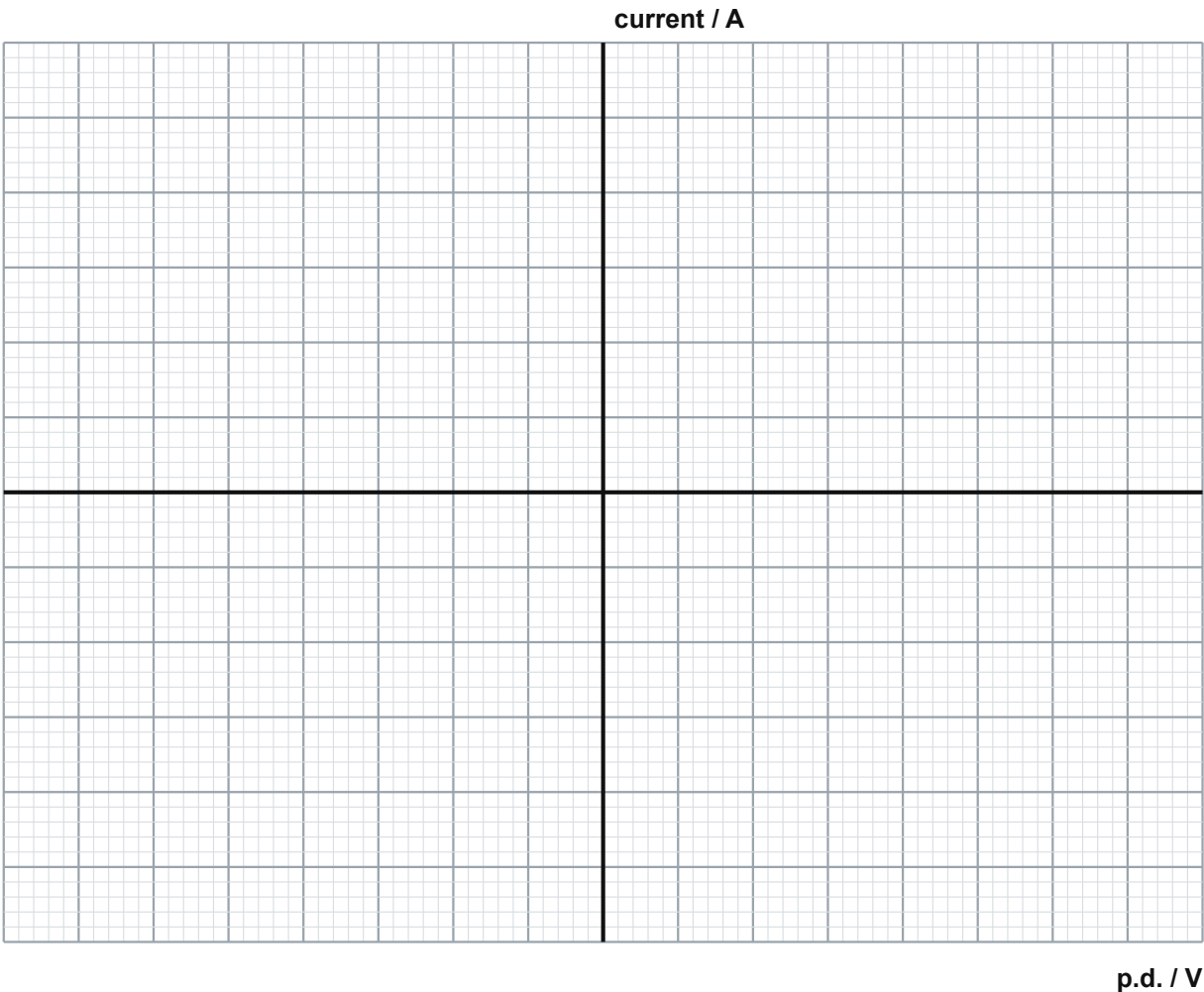
**Q2** Calculate the resistance of the fixed resistor using one pair of your readings.

Equation .....  
Substitute .....  
Answer ..... unit .....

# Analysis   Graphs and conclusions

## 2   Filament lamp graph

Plot current against p.d. for the filament lamp. Add scales to both axes. Draw a smooth curve of best fit.

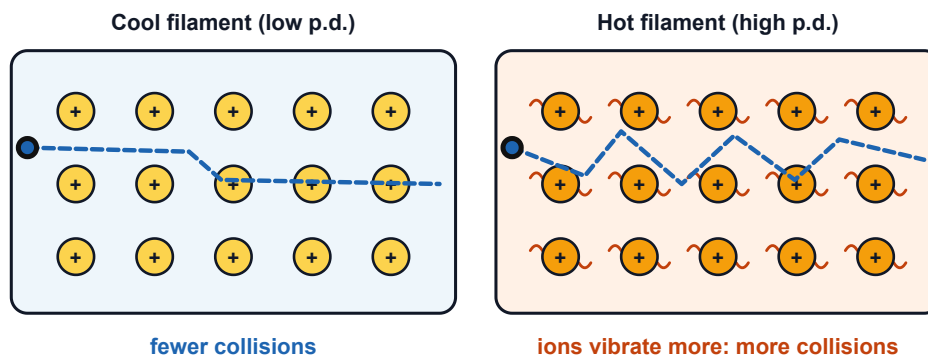


**Q3** Use your readings to calculate the resistance of the lamp at your lowest and highest p.d. What happens to the resistance?

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## Analysis Graphs and conclusions



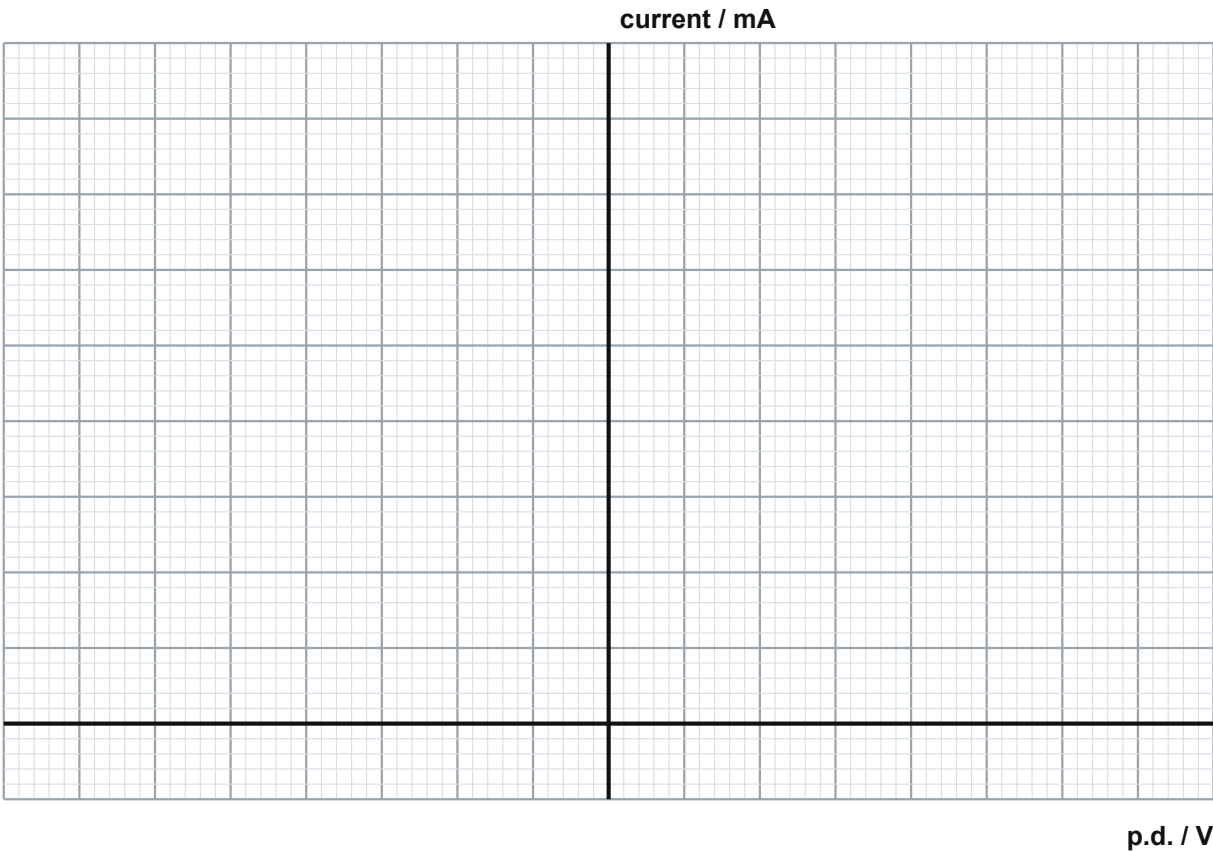
Electrons (blue path) collide with the metal ions (+). A hotter filament has vibrating ions, so there are more collisions and a higher resistance.

**Q4** Explain why the resistance of the filament lamp changes. **[3 marks]**

# Analysis   Graphs and conclusions

## 3   Diode graph

Plot current against p.d. for the diode. Add scales to both axes. Draw a smooth curve of best fit.



**Q5** Describe the I–V graph for the diode in the forward and reverse directions.

**Q6** Explain why a protective resistor is connected in series with the diode.

**Q7** Suggest why smaller steps in p.d. were used for the diode than for the other components.

# Analysis    Graphs and conclusions

## 4    Compare the three components

Complete the table using your three graphs.

| Component      | Shape of the I–V graph | As the p.d. increases, the resistance... | Ohmic? |
|----------------|------------------------|------------------------------------------|--------|
| Fixed resistor |                        |                                          |        |
| Filament lamp  |                        |                                          |        |
| Diode          |                        |                                          |        |

**Q8** A lamp has a p.d. of 4.5 V across it and a current of 0.26 A through it. Calculate its resistance.

Equation

Substitute

Answer

unit

**Q9** Why were readings also taken with the supply reversed?

**Q10** At the highest p.d. the fixed resistor felt warm. Suggest how this would affect the top end of its I–V graph.

# Exam practice Plan the practical

Exam practice • I–V characteristics • A six-mark method question like the ones AQA asks about this practical. Write your answer without looking at the method.

**AQA-STYLE** **Q11** Describe a method to find how the current through a filament lamp depends on the p.d. across it. Include both positive and negative values of p.d. You may include a circuit diagram in your answer. **[6 marks]**

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## 1 Check your answer

When you have finished, tick each point your method includes.

- |                                                                                   |                                                                                     |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <input type="checkbox"/> A circuit: lamp, ammeter and variable resistor in series | <input type="checkbox"/> Record the current at each p.d.; stay below the rated p.d. |
| <input type="checkbox"/> Voltmeter connected across the lamp only                 | <input type="checkbox"/> Reverse the supply and record negative values              |
| <input type="checkbox"/> Change the p.d. with the variable resistor, in steps     | <input type="checkbox"/> Plot current (y) against p.d. (x); draw a smooth curve     |